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physiological processes like respiration, digestion and excretion. The bulk flow of fluids can transport substances to long distances faster than by diffusion. The human circulatory system can circulate a millilitre of blood from the heart to feet and back again within 60 sec, rather than 60 years which may be needed if it were by diffusion.

Within our body the transport system helps in the coordination of physiological processes by transporting chemical signals from one place to another and assisting in the defence of the body by transporting immune cells to the sites of infection. These processes contribute to overall homeostasis (maintenance of constant internal environment). Movement of respiratory gases, hormones, nutrients, wastes and heat are carried by the circulatory system as shown in Figure 7.1.

Oxygen and carbon dioxide are exchanged in the lungs and tissues whereas nutrients from the digestive system are carried to the liver and the wastes from the tissues are carried by the blood and finally removed by the kidneys. The hormones are transported to their target organs. Circulatory system helps to maintain the homeostasis of the body fluids and body temperature (heat exchange).

The homeostatic regulation of the cardio vascular system maintains blood flow, or perfusion, to the heart and brain. In vasovagal syncope (fainting), signals from the nervous system cause a sudden decrease in blood pressure, and the individual faints from lack of oxygen to the brain.

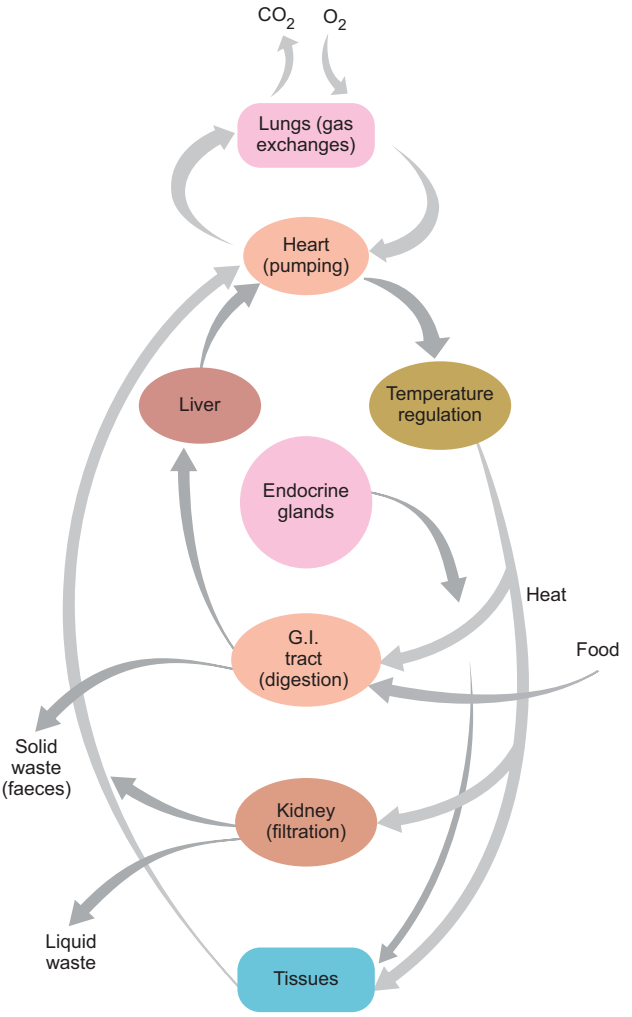

Figure 7.1 Schematic representations of the major functions of the circulatory system

In this chapter you will learn how the heart and blood vessels work together most of the time to prevent such problems.

7.1 Body fluids

The body fluid consists of water and substances dissolved in them. There are two types of body fluids, the intracellular fluid present inside the cells and the extracellular fluid present outside the cells. The three types of extracellular fluids are the **interstitial fluid** or tissue fluid (surrounds